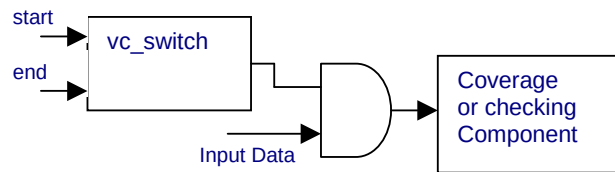


VC_SWITCH

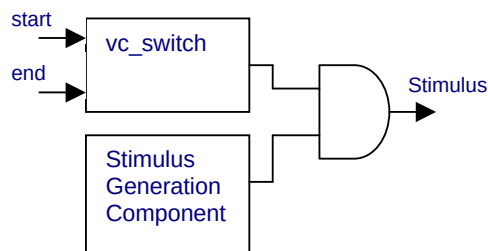
Overview

vc_switch allows enabling and disabling of various verification components upon reception of the specified events. It can be used to define time periods for the checking, coverage collection, stimulus generation, etc. Two inputs define the start event (switch on) and end event (switch off), correspondingly. The following figure illustrates the use of the vc_switch component:

Coverage & Checking Control



Stimulus Generation Control



Syntax

```
vc_switch #(start_event, end_event) Inst_name (clk, rst_n, start_in, end_in, switch_out);
```

Interface	
Clk	Clock
rst_n	Active-low reset
start_in	One-bit data input ; turns the switch on
end_in	One-bit data input ; turns the switch off
switch_out	One-bit output; presents the status of switch
Parameters	
Start_event	These parameters define a switch toggling event for turning the switch on (start_event) and off end_event). Events are enumerated in the following way: 0 - rising edge 1 - falling edge 2 - change (any edge) 3 - level The default value for both the start_event and end_event is 0 (rising edge).
End_event	

Example

```
module test;

reg a_reg;
reg clk, rst_n;
wire switch_out;

initial begin
    clk = 1;
    rst_n = 0;
    a_reg = 0;
end

always #5 clk <= ~clk;

vc_switch #(0,0) a_switch (clk, rst_n, a_reg, a_reg, switch_out);

initial begin
    $monitor ("[%0t]: a_switch = %b", $time, switch_out);
    #25 rst_n = 1;
    #40 a_reg = 1;
    #10 a_reg = 0;
    #40 a_reg = 1;
    #10 a_reg = 0;

    #40 $finish;
end

endmodule
```